



Thep Watana Co., Ltd.

382 Bangpoo Industrial Estate Sukhumvit Rd., Preaksa, Muang District, Samutprakarn 10280, THAILAND

Phone: +66-2721-3510

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SAFETY DATA SHEET

Company Identification

Thep Watana Co., Ltd. Address: 382 Bangpoo Industrial Estate.

Sukhumvit Road, Preaksa, Muang District, Samutprakarn 10280, THAILAND

Phone: +66(0)-2721-3510;

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1. CHEMICAL IDENTIFICATION

Product Identification

Trade name: Paco

Product name: paclobutrazol 10% WP

Product use: Plant growth regulator

Relevant identified uses of the substance/mixture and uses advised against:

Used as a plant hormone gibberellin to inhibit gibberellin biosynthesis, reducing internodal growth to give stouter stems, increasing root growth, causing early fruitset and increasing seedset in plants such as tomato and pepper. PBZ has also been shown to reduce frost sensitivity in plants. Moreover, paclobutrazol can be used as a chemical approach for reducing the risk of lodging in cereal crops. PBZ is used by arborists to reduce shoot growth and has been shown to have additional positive effects on trees and shrubs. Among those are improved resistance to droughtstress, darker green leaves, higher resistance against fungi and bacteria, and enhanced development of roots. Cambial growth, as well as shoot growth, has been shown to be reduced in some tree species.

2. HAZARD IDENTIFICATION

Potential Effects to environment

Dangerous for environment

In soil, the compound degraded much more rapidly, and did not accumulate in long-term studies (3-5 y).

3. COMPOSITION/INFORMATION ON INGREDIENT

Chemical name: (2*RS*,3*RS*)-1-(4-chlorophenyl)-4,4-dimethyl-2-(1*H*-1,2,4-triazol-1-yl)pentan-3-ol

Common name: paclobutrazol

Chemical family: Triazole



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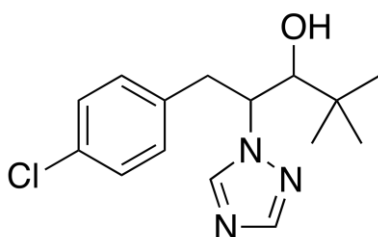
Ingredients and composition:

Component	CAS No.	R-sentences	Content
Active ingredient - paclobutrazol	76738-62-0	13	10%

Formula: $C_{15}H_{20}ClN_3O$

Molecular weight: 293.795 g/mol

Structural formula:



4. FIRST AID MEASURES

Description of first aid measures

In case of swallowed:

Give victim 1 or 2 glasses water to drink. Seek medical advice before inducing vomiting.
Never give food or drinks to an unconscious victim. In case of accident or if you feel unwell, seek medical advice (show the label where possible).

In case of eyes contact:

Immediately flush with water for 15 minutes, occasionally lifting the upper and lower lids.
If irritation persists, seek medical attention.

In case of skin contact:

Wash immediately with plenty of water and soap. If necessary seek medical attention.

In case of Inhaled contact:

Move victim to fresh air. If needed, administer oxygen or apply artificial respiration. Obtain medical attention if necessary.

5. FIRE FIGHTING MEASURES

Suitable Extinguishing Media:

CO₂, dry powder, foam or water spray.

Unsuitable Extinguishing Media:

High-pressure water jet.



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Fire Fighting Equipment:

Wear an independent respirator.

Specific Method:

When possible, let the fire burn out because the amount of water can enlarge the contaminated surface. Cool containers / tanks with water.

6. ACCIDENTAL RELEASE MEASURES (SPILLAGE)

Personal precaution measures:	use suitable protective agents.
Environmental protection measures:	do not allow entering drains or watercourses.
Cleaning, clearing measures:	take up mechanically. Clean up affected area. Use an adequate container for disposal.

7. HANDLING AND STORAGE

Good industrial practice in housekeeping and personal hygiene should be followed.

Safe Handling:

Handle product in a well-ventilated area. Normal precautions for handling chemicals should be respected. Use personal protective agents. Keep away from heat or ignition sources.

Storage Conditions:

Store in a cool, dry, well-ventilated area, out of reach of children. Store in fireproof location. Keep package well closed. Do not store together with strong acids and alkali. Keep away from food, drinks and animal feeding stuff.

8. EXPOSURE CONTROL/PERSONAL PROTECTION

Measurement

Technical measures:	provide adequate ventilation, especially in closed areas.
Hygienic measures:	wash hands before each break and after work. Keep away from food, drinks and animal feeding stuff. Do not eat, drink or smoke during handling.

Personal Protective Equipment

Eyes Protection:

Safety goggles.

Respiratory Protection:

Use a respirator with a P1 filter.

Protective Material Types:

Protective gloves.



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9. PHYSICAL AND CHEMICAL PROPERTIES

Information on basic physical and chemical properties

Physical stage:	Loose powder
Color:	Off-white
Odor:	Weak
pH:	6 – 8 @ 20°C
Relative density:	Not applicable

10. STABILITY AND REACTIVITY

Stability:	stable under normal, dry storage conditions.
Conditions to avoid:	formation of dust in closed areas.
Materials to avoid:	acid and alkali conditions
Hazardous decomposition products:	SO ₂ and NO _x

11. TOXICOLOGICAL INFORMATION

Information on Acute toxicity for end product, Paclobutrazol:

Not available

Information on local effect:

Skin irritation (rabbit):	Not irritant.
Eye irritation (rabbit):	Not irritant.

12. ECOLOGICAL INFORMATION

Not available

13. DISPOSAL CONSIDERATIONS

Remainders of spraying solution ought to be diluted 10 times and sprayed over the treated plot according to the descriptions of use. Do not pollute ponds, watercourses and canals with the product or the empty packaging. Never reuse the empty packaging for other materials. To avoid spraying solutions after treatments, calculate how much liquid is needed on the basis of the area that need to be treated and the amount of water needed for each hectare

The empty packaging need to be rinsed with water by means of a manual system (three successive times shaking with water) or by means of a cleaning system with water under pressure, placed on



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the spraying tank. The wash water needs to be poured in the spraying tank. The rinsed packaging must be brought to a specific collector.

14. TRANSPORT INFORMATION

The data provided in this section is for information only. Please apply the appropriate regulations to properly classify your shipment for transportation.

UN No:	3082
Dangerous Goods Class:	9
Packaging Group:	III
Marine pollutant:	Yes
DOT/IMO Proper Shipping name:	NOT REGULATED

15. REGULATORY INFORMATION

Risk Phrases:

R50/53: Very toxic to aquatic organisms; may cause long-term adverse effects in the aquatic environment.

Safety Phrases:

S2: Keep out of reach of children.

S13: Keep away from food, drink and animal feeding stuffs.

S20/21: When using, do not eat, drink or smoke.

16. OTHER INFORMATION

All information and instructions provided in this SDS are based on the current state of scientific and technical knowledge at the date indicated on the present SDS and are presented in good faith and believed to be correct. This information applies to the PRODUCT AS SUCH. In case of formulations or mixes, it is necessary to ascertain that a new danger will not appear. It is the responsibility of persons in receipt of this MSDS to ensure that the information contained herein is properly read and understood by all people who may use, handle, dispose or in any way come in contact with the product.